Design Proposal: Mach-Zehnder Interferometer

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Abstract: This study investigates the performance of single-mode Silicon MZIs (500×220 nm) under fundamental TE polarization and TM. By implementing path-length imbalances (ΔL) of 40, 80, 160, and 250 μm, we demonstrate a systematic approach to extracting waveguide properties. The following sections present the MZI theory used in this study, the simulation parameters, and a detailed verification of the theory using experimental data extracted from fabricated devices.

1. Introduction

Modern optical communication is strongly linked with silicon photonics, being a field that is benefiting from the infrastructure of CMOS foundries that are becoming more and more mature and also capable of achieving a higher production volume [1].

It is seen that silicon photonics enables the integration of optical systems in single and also compact chips [2]. From the several existent photonics systems, the Mach-Zehnder Interferometer (MZI) is seen as an indispensable component, mainly due to it versatility, to be used from signal modulation to wavelength filtering [3].

The current work details the design of several MZI configurations, targetting to measure waveguide group indices using an experimental setup. These designs are optimized for E-beam lithography to guarantee reliable fabrication. In this work, it is provided an overview of the MZI theory, component-level modeling, and the design variations.

2. Theory

In silicon photonics, it is possible to find several types of fundamental blocks. In this study, it will be introduced a Mach-Zehnder interferometer with 2 Y-branches, as shown in Fig. 1.

The MZI block operates by splitting the light in 3-dB, using a Y-branch, into two waveguides and then bringing them together into a single output, by using again another Y-branch. By making the lengths of the waveguides (being the branches of the interferometer) having different values, such as L_1 and L_2, we make this system unbalanced, triggering the interference of the field distribution, in the system's waveguide.

The expected output, observed in MZI systems are sinusoidal variations caused by the length imbalance, defined as free spectral range. The mathematical deduction will be done in the following sessions [3].

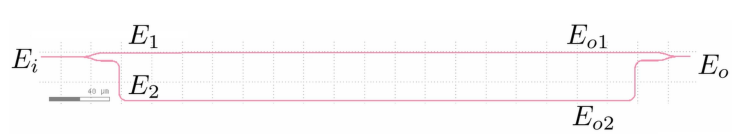


Fig. 1. Conceptual setup of an unbalanced Mach-Zehnder interferometer.

2.1. Theoretical Framework

This section introduces the basic equations to model a simple MZI. Equation 1 shows the propagation of light inside the waveguide.

$$\beta_{1,2} = \frac{2\pi n_{1,2}}{\lambda} \tag{1}$$

As observed in 1, at the end of each branches, the fields are described by equation 2 and 3. In these expressions, α_1 and α_2 represent the power attenuation coefficients.

$$E_{o1} = \frac{E_i}{\sqrt{2}} \exp\left(-i\beta_1 L_1 - \frac{\alpha_1}{2} L_1\right) \tag{2}$$

$$E_{o2} = \frac{E_i}{\sqrt{2}} \exp\left(-i\beta_2 L_2 - \frac{\alpha_2}{2} L_2\right) \tag{3}$$

At the second Y-junction, on its output, the total output field E_o can be described as:

$$E_o = \frac{1}{\sqrt{2}}(E_{o1} + E_{o2}) = \frac{E_i}{2}\left[\exp\left(-i\beta_1 L_1 - \frac{\alpha_1}{2} L_1\right) + \exp\left(-i\beta_2 L_2 - \frac{\alpha_2}{2} L_2\right)\right] \tag{4}$$

Since the optical intensity I is proportional to the square of the electric field magnitude ($I \propto |E|^2$), the output intensity I_o is given by:

$$I_o = \frac{I_i}{4}\left|\exp\left(-i\beta_1 L_1 - \frac{\alpha_1}{2} L_1\right) + \exp\left(-i\beta_2 L_2 - \frac{\alpha_2}{2} L_2\right)\right|^2 \tag{5}$$

It is possible to simplify this model by assuming nearly lossless waveguides ($\alpha_1 = \alpha_2 \approx 0$), and given that the waveguides in both branches are structurally identical, it results in the same effective index ($n_1 = n_2 = n_{\text{eff}}$) and propagation constant ($\beta_1 = \beta_2 = \beta$).

As a result, the interference found on this model, it is just linked to the physical path length difference, $\Delta L = L_2 - L_1$, as despicted in equation 6.

$$I_o = \frac{I_i}{2}\left[1 + \cos(\beta \Delta L)\right] \tag{6}$$

Another way to understand equation 6 is through the normalized transfer function:

$$\frac{I_{\text{out}}}{I_{\text{in}}} = \cos^2\left(\frac{\Delta\phi}{2}\right) \tag{7}$$

where the total phase difference $\Delta\phi$ is defined as:

$$\Delta\phi = \beta \Delta L = \frac{2\pi n_{\text{eff}}}{\lambda} \Delta L \tag{8}$$

The periodic nature of Eq. 7 results in a transmission spectrum characterized by distinct interference fringes. The waveguide's group can be extracted by using thoses spectral peaks and nulls [3].

2.2. Free Spectral Range (FSR) and Group Index Extraction

As described at the beginning of this session, from the sinosoidal functions, observed in the output of the MZI system, it is possible to extract the FSR (Free Spectral Range), a key metric used for characterizing the device's spectral response.

For an imbalanced MZI, the FSR is a function of both the physical path imbalance (ΔL) and the waveguide group index (n_g) [3], as seen in equation 9.

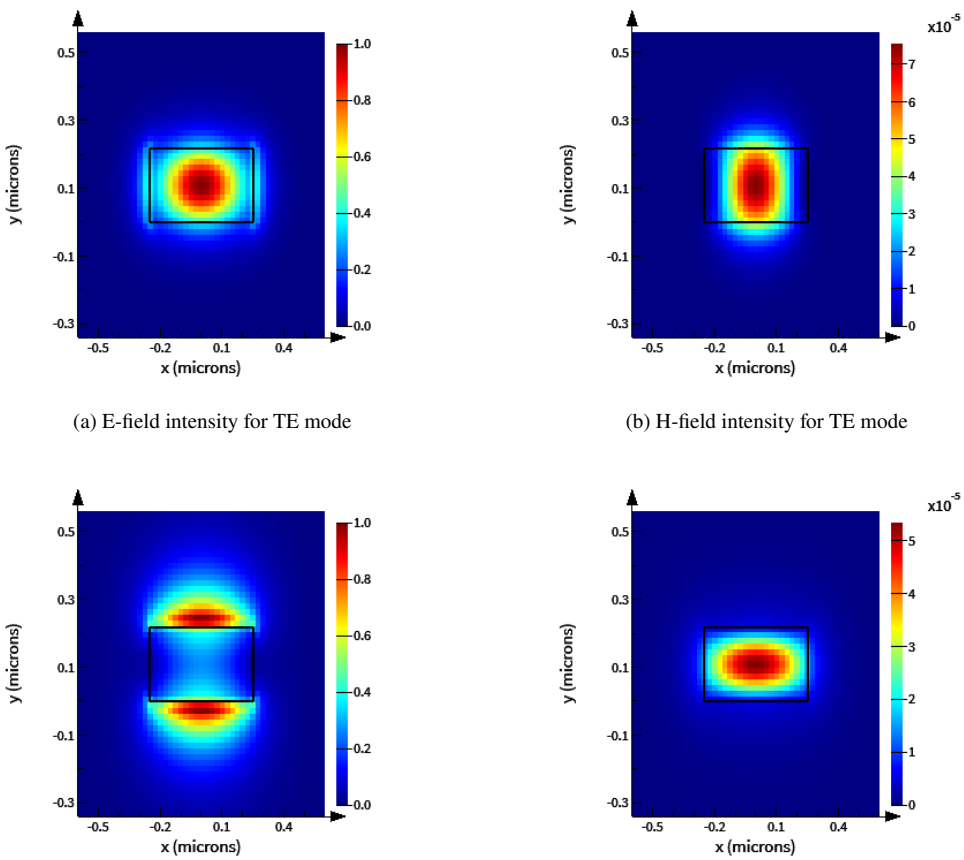
$$FSR = \frac{\lambda^2}{n_g \Delta L} \tag{9}$$

It is important to highlight that n_g is more important than the standard refractive index (n_{eff}) because the FSR depends on how the entire pulse is delayed between the two branches, in our MZI. If instead n_{eff} is used to calculate the Free Spectral Range, at the end the results would be incorrect. Due to the dispersive behavior in silicon, the phase velocity suffer fast variations that it "pulls" or "pushes" the group envelope, resulting in changes in the effective spaces between the observed interference peaks. [3] **any other ref???**.

3. Modelling and Simulation

3.1. Waveguide Design and Modal Analysis

The devices use a standard **strip waveguide** architecture. It is used a fixed **220 nm** silicon thickness and a waveguide width of **500 nm**. This geometry is a standard industry benchmark for the C-band, as it has high modal confinement and also supports single-mode operation for the fundamental **quasi-TE polarization** and **quasi-TM polarization**, shown in Fig. 2.



(a) E-field intensity for TE mode

(b) H-field intensity for TE mode

(c) E-field intensity for TM mode

(d) H-field intensity for TM mode

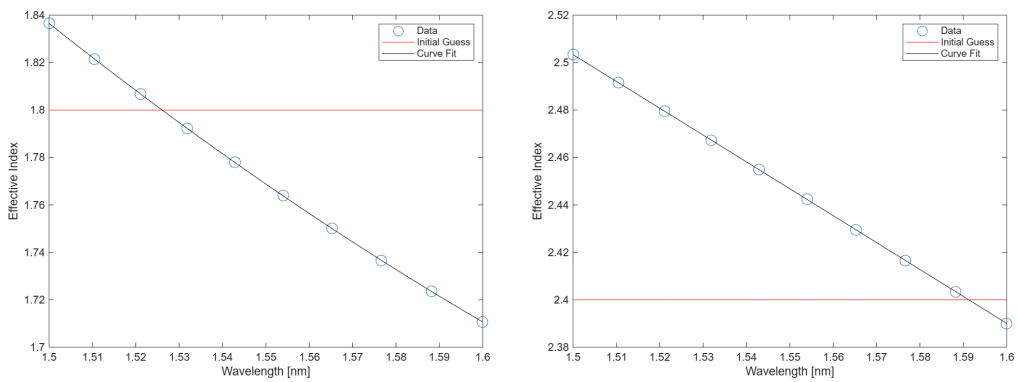
Fig. 2. Mode profiles obtained by using eigensolver analysis [4], for a silicon strip waveguide of 500×220 nm.

The simulation also shows the optical confinement, achieved by the silicon core, for both TE and TM modes.The quasi-TE mode shows higher confinement inside the silicon core, however, the quasi-TM mode has a larger field dispersion extending into the cladding, as presented in Fig. 2a and Fig. 2c, respectively.

To accurately characterize the modal properties of the given geometry, numerical simulations are performed using an eigenmode solver [4], for the given waveguide cross-section. From that, it is possible to extract of the effective index and dispersion characteristics, with higher accuracy, for the MZI designs. For TE mode the effective index obtained is $n_{eff} = 2.447$ and for TM mode, the effective index obtained is $n_{eff} = 1.769$.

Based on the second-order Taylor expansion, the effective index can be represented by Eq. 10.

$$n_{\text{eff}}(\lambda) = n_{\text{eff1}} + n_{\text{eff2}}(\lambda - 1.55) + n_{\text{eff3}}(\lambda - 1.55)^2 \tag{10}$$



(a) Effective index for TE mode is given by $n_{\text{eff}}(\lambda) = 2.447 + -1.133(\lambda - 1.55) + -0.044(\lambda - 1.55)^2$

(b) Effective index for TM mode is given by $n_{\text{eff}}(\lambda) = 1.769 + -1.259(\lambda - 1.55) + 1.913(\lambda - 1.55)^2$

Fig. 3. Curve fitting for effective index in function of wavelength.

Table 1 shows all coefficients obtained after running the numerical simulation, for both TE and TM. Fig. 5 display the curve fitting using the coefficients.

Table 1. Second-order Taylor coefficients for the effective index (n_{eff}) of TE and TM modes at $\lambda_0 = 1.55$ µm with 6-decimal precision.

Mode	n_{eff1}	n_{eff2} (µm^{-1})	n_{eff3} (µm^{-2})
Quasi-TE	2.446821	-1.133395	-0.043934
Quasi-TM	1.768821	-1.258670	1.913483

3.2. Circuit Simulation Setup

The circuit was modeled using the Lumerical INTERCONNECT circuit simulator, with the configuration depicted in Fig. 4. This system-level simulation uses the dispersive waveguide properties calculated before, and also S-parameter models for the grating couplers and Y-branch elements, also included in the simulation. This ensure high accuracy to predict the spectral behavior of the device that will be fabricated.

The spectral response of the input and output grating couplers is shown in Fig. 5a. The data show a Gaussian-like profile, and its peak transmission is centered around 1550 nm, being the target C-band wavelength range for this MZI project.

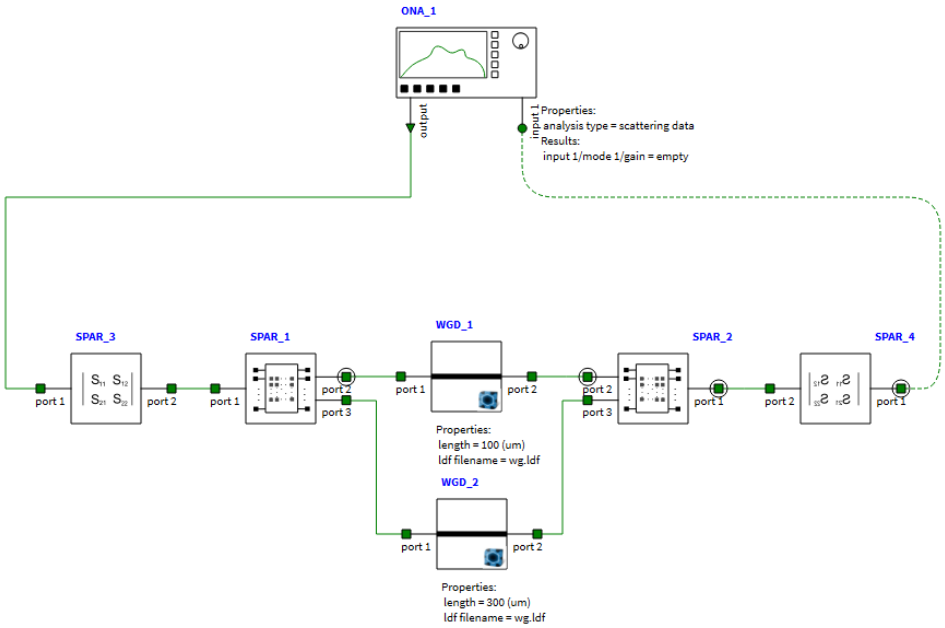
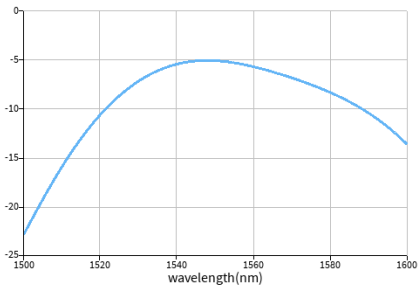
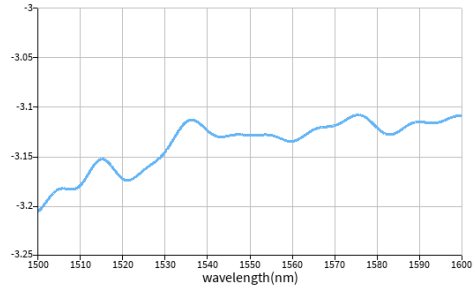


Fig. 4. System-level circuit created on Lumerical INTERCONNECT environment. The schematic illustrates the integrated simulation, in which waveguide segments are connected with S-parameters of the Y-junctions (named as SPAR_1 and SPAR_2) and grating couplers (named as SPAR_3 and SPAR_4), to obtain the full spectral response of the MZI.



(a) Simulated transmission characteristics for a back-to-back grating coupler.



(b) Simulated transmission spectrum of the Y-branch splitter, demonstrating broadband 3-dB power splitting.

Fig. 5. Spectral response from 1500 to 1600 nm.

The spectral response for the Y-branch splitter is presented in Fig. 5b. It is possible to see a stable loss of approximately 3 dB across the C-band, showing that the input power is equally divided betzeen the 2 outputs, with minimal parasitic loss.

3.3. MZI Design Variations

The designs for the MZIs was executed on KLayout software [5]. Some classical MZIs with different length imbalances were implemented. Besides that, a constant waveguide cross-section and identical Y-junctions were used in the designs, and TE and TM polarizations were also

explored. Fig. 6 shows the first layout draft with 3 different MZIs.

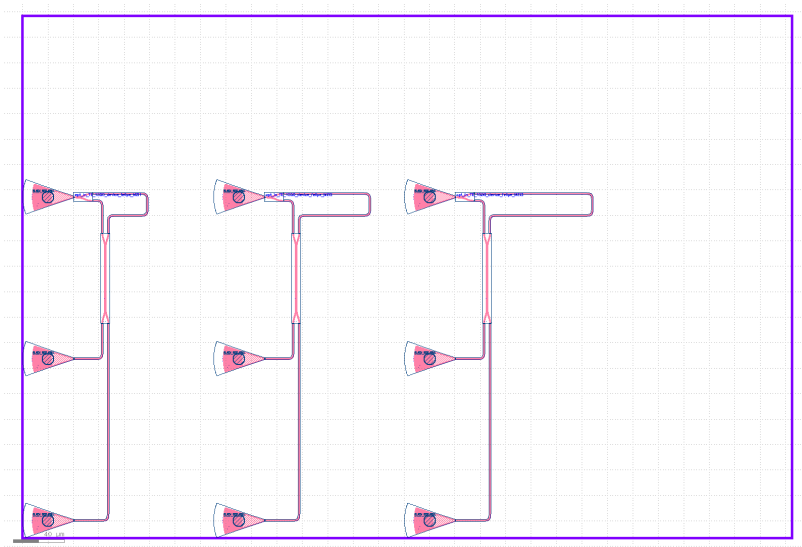


Fig. 6. First draft setup, using KLayout mask design for the three MZI variants.

The final layout is shown in Fig. 7. Table 2 summarizes the geometric parameters for each design, with its respective predicted FSR, calculated at a center wavelength of 1550 nm. For designs using TE mode, the simulated group index is $n_g \approx 4.197$, whereas for TM mode, the group index is $n_g \approx 3.75$.

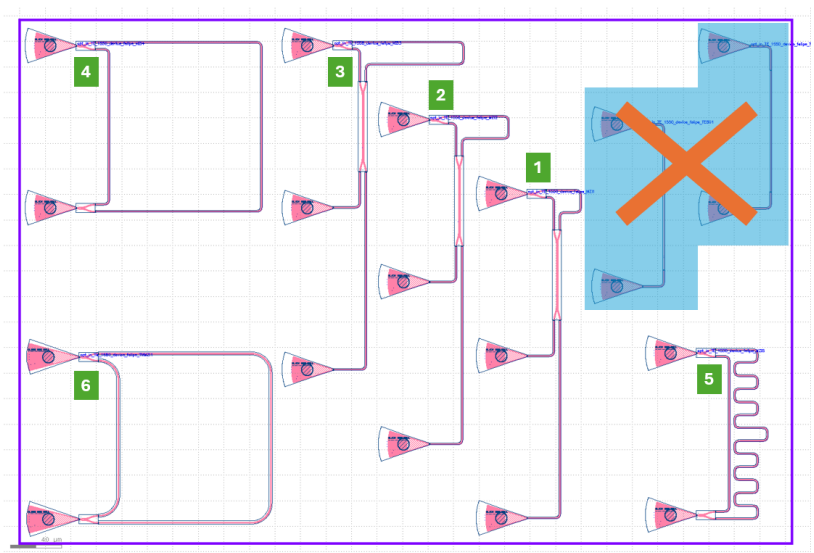


Fig. 7. Final KLayout mask designs for 6 MZI variants. Each design used in this study is numbered, and the blue region shows layouts that will not be taken into consideration, as they are not an MZI design.

The designs 4, 5, and 6 have the same ΔL. The variation to be studied are the waveguide serpentines influence, by making comparisons between designs 4 and 5, and a comparison

Table 2. MZI Design Parameters and Predicted Free Spectral Range

Design #	Width	Mode	L_1 (µm)	L_2 (µm)	ΔL (µm)	Expected FSR (nm)
1	500 nm	TE	30	70	40	14.30
2	500 nm	TE	30	110	80	7.15
3	500 nm	TE	30	190	160	3.58
4	500 nm	TE	140	390	250	2.29
5	500 nm	TE	140	390 (waveguide serpentines)	250	2.29
6	500 nm	TM	140	390	250	2.56

between TE and TM modes, by analyzing the result of designs 4 and 6.

4. Fabrication

The photonic integrated circuits (PICs) were fabricated using the NanoSOI MPW process, done by Applied Nanotools Inc. [6] The technical specifications and process used in this project are summarized below:

Substrate and Material: The devices were processed on 200 mm SOI wafers. Besides that, the silicon layer height is 220 nm, and it is buried in a oxide (BOX) layer of 2 µm. Later, the wafer is diced into 25 × 25 mm substrates.

Electron Beam Lithography (EBL): The patterns were created by a JEOL JBX-8100FS 100 keV EBL system.

Etching and Development: Once the exposure is complete, the patterns are developed using a solution called tetramethylammonium sulfate (or TMAH). The different shapes of the circuit are then etched into the silicon by an isotropic ICP-RIE (Inductively Coupled Plasma Reactive Ion Etching).

Cladding and Metrology: At the end, a 2.2 µm thick SiO_2 upper cladding is deposited using plasma-enhanced chemical vapor deposition (or PECVD) at 300°C. After, it is verified the layer thicknesses and etch quality by using mainly the Scanning Electron Microscopy (SEM) method.

However, the fabrication process can cause a mismatch among devices, generating variability in both the width and height of the waveguide. This work uses, as reference, the variation provided by Soitec, France.

- Wafer diameter: 6";
- Si height: average size of 219.2nm, with a σ deviation of ± 3.9nm;
- Si width: 500 +10/-30nm;

Those information are important to execute the corner analysis of the optical device. Besides the nominal dimension, it is also simulated 4 different corners, as shown in Fig. 8, using the deviations for the waveguide.

Based on the corners diamensions for the device's waveguide, it was possible to obtain the effective index (n_{eff}), and the group index (n_g) at λ_0 = 1550 nm, as seen in Tables 3 and 4.

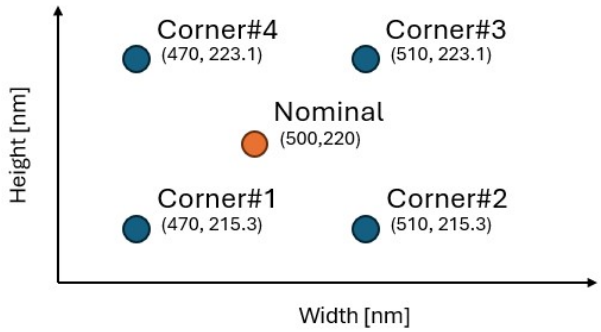


Fig. 8. Corner analysis representation, displaying variation in both width and height, from the waveguide.

Table 3. Corner Analysis for **TE Mode**: Effective Index (n_{eff}), Polynomial Models, and Group Index (n_g) at λ_0 = 1550 nm.

Corner	n_{eff} **at** 1.55 **µm**	**Taylor Expansion Polynomial** ($n_{\text{eff}}(\lambda)$)	n_g
Nominal	2.4468	$2.4468 - 1.1334(\lambda - 1.55) - 0.0439(\lambda - 1.55)^2$	4.2036
Corner 1	2.3725	$2.3725 - 1.2132(\lambda - 1.55) - 0.0341(\lambda - 1.55)^2$	4.2529
Corner 2	2.4416	$2.4416 - 1.1213(\lambda - 1.55) - 0.0333(\lambda - 1.55)^2$	4.1796
Corner 3	2.4722	$2.4722 - 1.1068(\lambda - 1.55) - 0.0465(\lambda - 1.55)^2$	4.1878
Corner 4	2.4039	$2.4039 - 1.1989(\lambda - 1.55) - 0.0528(\lambda - 1.55)^2$	4.2621

Table 4. Corner Analysis for **TM Mode**: Effective Index (n_{eff}), Polynomial Models, and Group Index (n_g) at λ_0 = 1550 nm.

Corner	n_{eff} **at** 1.55 **µm**	**Taylor Expansion Polynomial** ($n_{\text{eff}}(\lambda)$)	n_g
Nominal	1.7688	$1.7688 - 1.2587(\lambda - 1.55) + 1.9135(\lambda - 1.55)^2$	3.7198
Corner 1	1.7167	$1.7167 - 1.1390(\lambda - 1.55) + 2.0469(\lambda - 1.55)^2$	3.4821
Corner 2	1.7443	$1.7443 - 1.1889(\lambda - 1.55) + 1.9518(\lambda - 1.55)^2$	3.5870
Corner 3	1.7963	$1.7963 - 1.3151(\lambda - 1.55) + 1.8184(\lambda - 1.55)^2$	3.8346
Corner 4	1.7664	$1.7664 - 1.2747(\lambda - 1.55) + 1.9529(\lambda - 1.55)^2$	3.7422

For each MZI layout, it was also calculated the predicted FSR, also considering the corner variations. Table 5 display the maximum and minimum values expected to be found in the measurements, obtained from the fabricated devices.

Table 5. Predicted FSR (nm) across Manufacturing Corners for MZI Designs 1–6 (Min: Corner 2, Max: Corner 4).

Design #	ΔL (µm)	Mode	Nominal FSR	Min FSR (C2)	Max FSR (C4)	Variation (ΔFSR)
1	40	TE	14.289	14.092	14.370	0.278
2	80	TE	7.144	7.046	7.185	0.139
3	160	TE	3.572	3.523	3.593	0.070
4	250	TE	2.286	2.255	2.299	0.044
5	250	TE	2.286	2.255	2.299	0.044
6	250	TM	2.583	2.506	2.760	0.254

5. Experiment data

The fabricated devices were characterized using a custom-built automated test station [7, 8]. The measurement process was controlled via a Python-based automated software framework [9].

An **Agilent 81600B** tunable laser served as the optical input source, while **Agilent 81635A** optical power sensors were utilized as output detectors. The laser wavelength was swept from 1500 nm to 1600 nm with a resolution step of 10 pm. To ensure precise polarization control, a **Polarization Maintaining (PM) fiber array** [10] was used for chip coupling. For the TE mode, light was coupled directly into the grating couplers; for TM mode characterization, a 90° rotation was applied to the input fiber to align the polarization with the TM-specific grating couplers [11].

The experimental phase involved the characterization of six Mach-Zehnder Interferometer (MZI) layouts.

Optical transmission spectra were collected for five Transverse Electric (TE) designs (Layouts 1–5), as shown in Fig. 9, which featured varying physical path differences (ΔL) ranging from 40 µm to 250 µm.

The raw spectral data for the TE modes showed high-extinction interference fringes, allowing for robust curve fitting. The measured Free Spectral Range (FSR) transitioned from approximately 14.3 nm for the shortest imbalance (40 µm) down to 2.3 nm for the longest imbalances (250 µm).

Layout 6 was designed for Transverse Magnetic (TM) mode operation. However, due to a fabrication error the TM mode did not yield measurable interference fringes. Consequently, experimental data is exclusively provided for the TE-polarized designs.

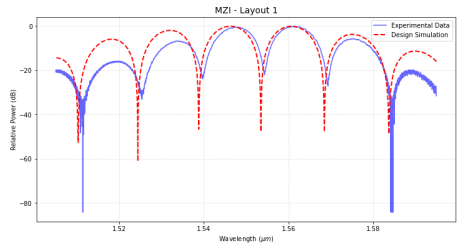
6. Analysis

The analysis focuses on comparing the extracted experimental parameters against the nominal and corner simulations (C1–C4) to determine the fabrication quality and model accuracy.

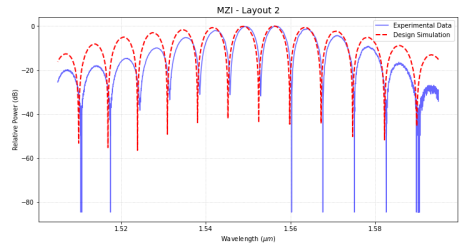
The extracted effective index (n_{eff}) and group index (n_g) were plotted against simulation boundaries.

The experimental values for n_{eff} at 1.55 µm remained consistently within the 2.42–2.44 range. This sits comfortably between the Minimum (C1) and Maximum (C3) simulation corners, validating the phase-velocity model of the waveguides.

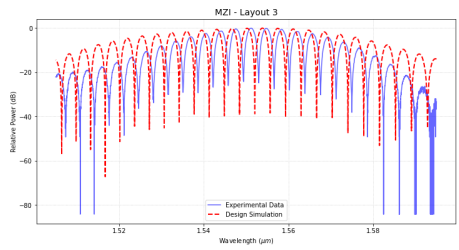
The group index values were found to be slightly lower than the nominal simulation (4.20). Since n_g is highly sensitive to the waveguide cross-section, this indicates a slight deviation in the



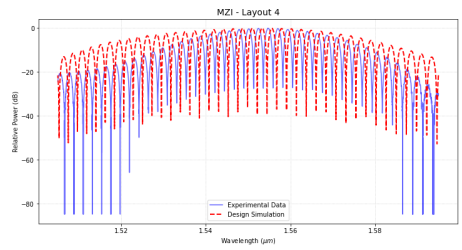
(a) Designed 1, MZI with TE mode with ΔL = 40 µm.



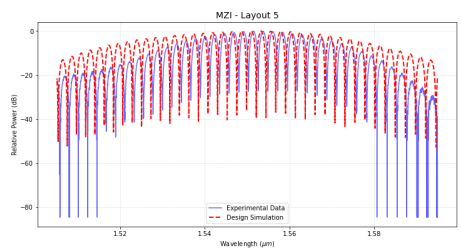
(b) Designed 2, MZI with TE mode with ΔL = 80 µm.



(c) Designed 3, MZI with TE mode with ΔL = 160 µm.



(d) Designed 4, MZI with TE mode with ΔL = 250 µm.



(e) Designed 5, MZI with TE mode with ΔL = 250 µm.

Fig. 9. Overlay of experimental data and simulated transfer functions for all MZIs designed, from 1 to 5.

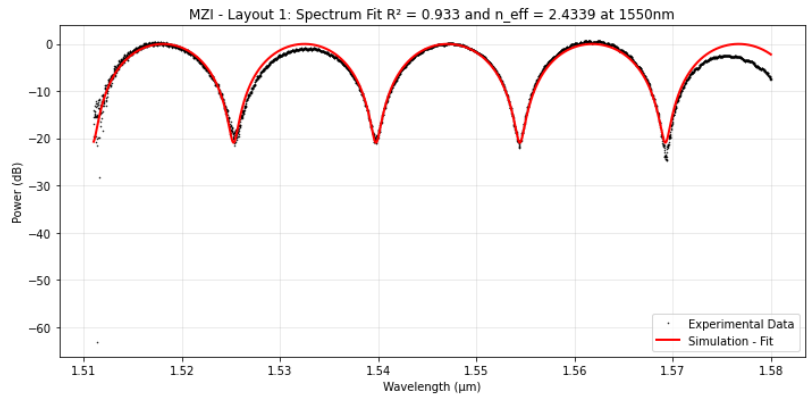
group velocity, which is further explored in the FSR analysis. The results are displayed in Fig. 10-14.

The FSR is the most critical metric for verifying the physical dimensions of the fabricated MZI. As shown in the Trellis plots and summary table, the experimental FSR values lean toward the Maximum Corner (C4). Table 6 display all FSR for each design variation.

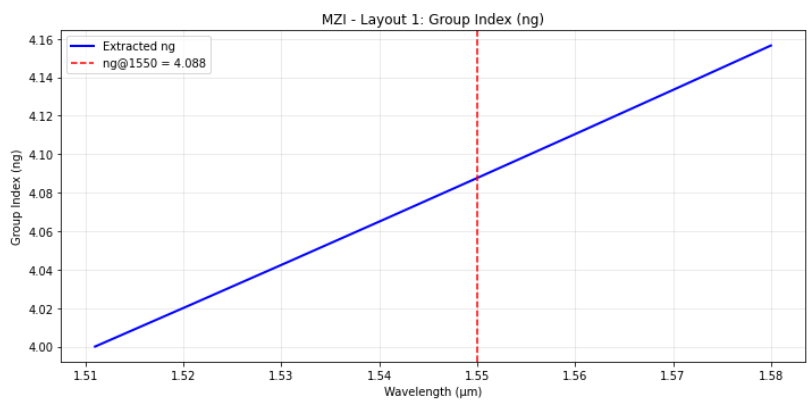
Besides that, the comparison between Layout 4 (straight) and Layout 5 (serpentine) for a fixed ΔL = 250 µm demonstrates the geometric scalability of the design. The Experimental FSR values are within 0.3% of each other, confirming that the serpentine folding does not introduce significant parasitic dispersion or deviations in the group index. This validates the use of compact serpentine delay lines for high-density MZI integration without compromising spectral accuracy.

7. Conclusion

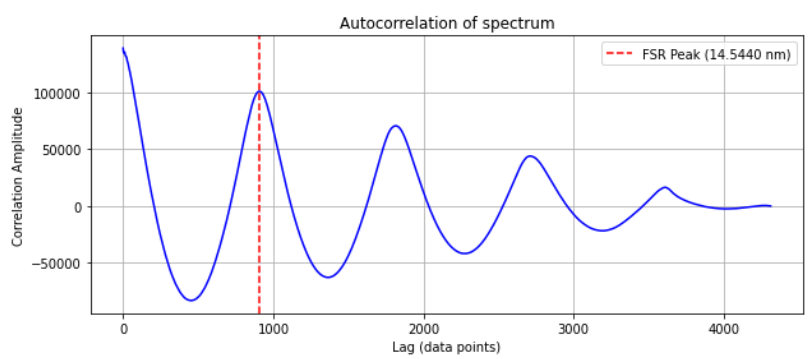
The experimental characterization of the Mach-Zehnder Interferometer (MZI) layouts 1–5 successfully validated the theoretical design and simulation models for TE-polarized light. While Design 6 (TM mode) was lost due to fabrication errors—a common occurrence given the higher



(a) Experimental data vs. fitted MZI transfer function.

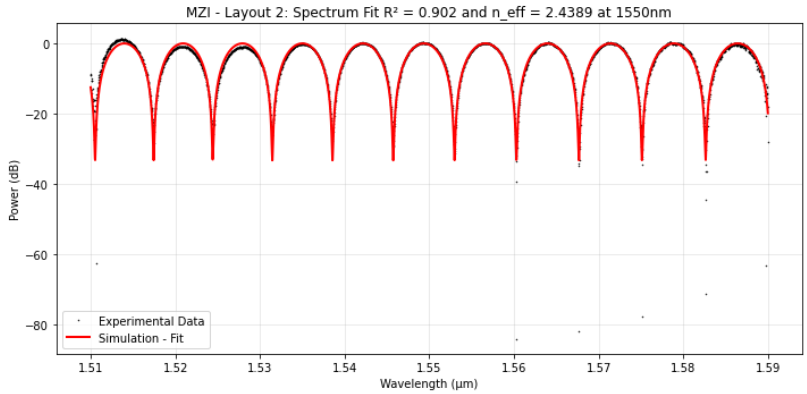


(b) Extracted group index n_g as a function of wavelength.

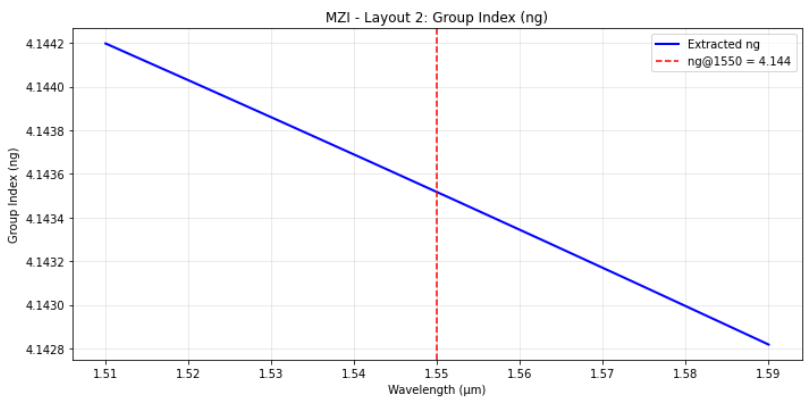


(c) Autocorrelation used to determine initial FSR and index shift.

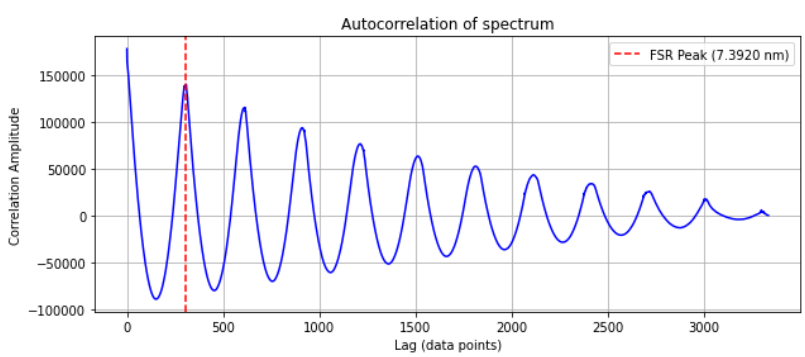
Fig. 10. Full analysis of MZI Layout 1. The effective index is modeled as $n_{eff}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2$, with extracted parameters: $n_1 = 2.4339$, $n_2 = -1.0669\,\mu m^{-1}$, and $n_3 = -0.7337\,\mu m^{-2}$. (a) Final numerical fit, (b) Group index extraction, and (c) Initial peak detection.



(a) Experimental data vs. fitted MZI transfer function for Layout 2.

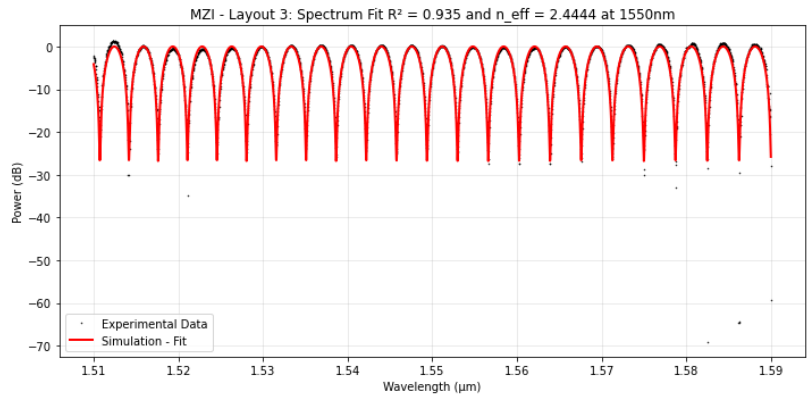


(b) Extracted group index n_g for Layout 2.

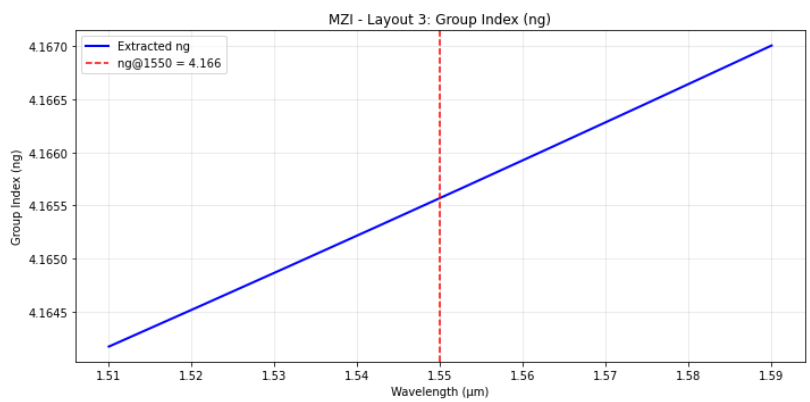


(c) Autocorrelation for Layout 2 used to determine initial FSR.

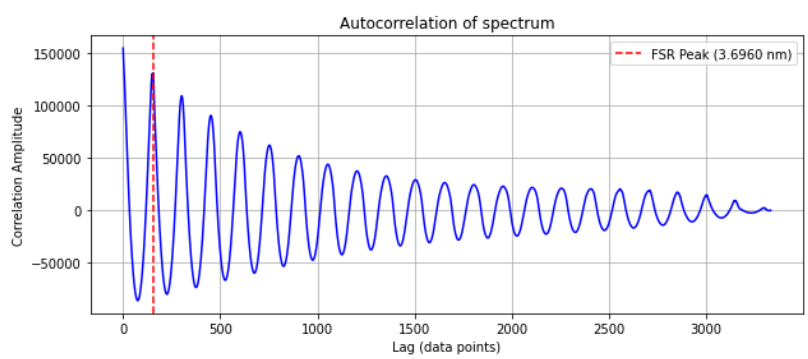
Fig. 11. Full analysis of MZI Layout 2. The effective index is modeled as $n_{eff}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2$, with extracted parameters: $n_1 = 2.4387$, $n_2 = -1.0999\,\mu m^{-1}$, and $n_3 = 0.0061\,\mu m^{-2}$. (a) Final numerical fit, (b) Group index extraction, and (c) Initial peak detection.



(a) Experimental data vs. fitted MZI transfer function for Layout 3.

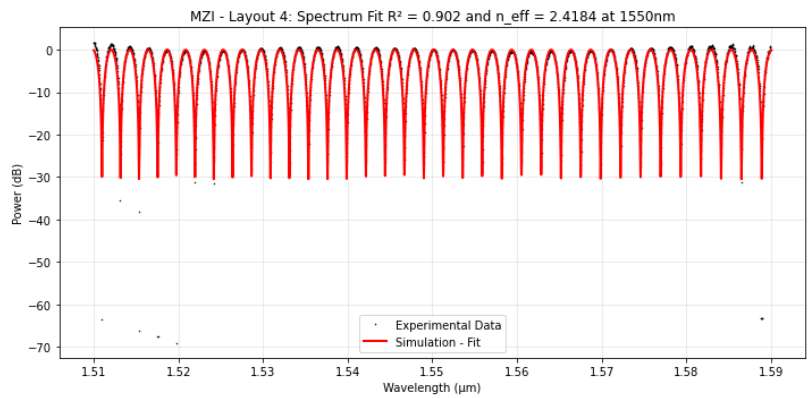


(b) Extracted group index n_g for Layout 3.

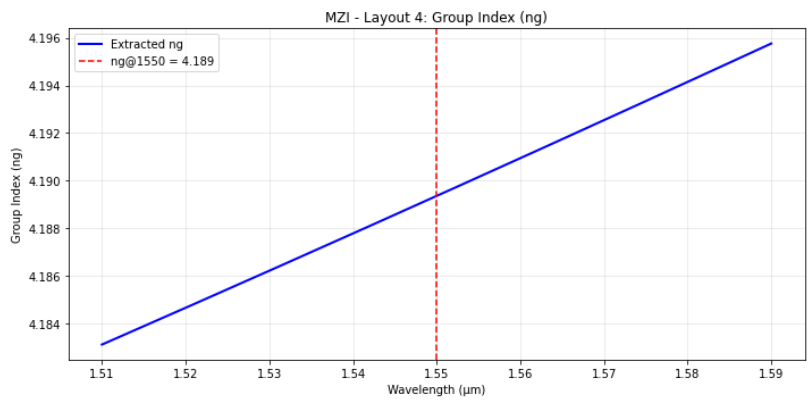


(c) Autocorrelation for Layout 3 used to determine initial FSR.

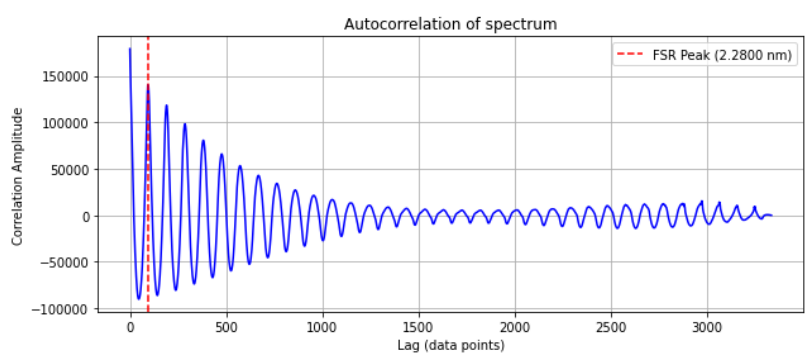
Fig. 12. Full analysis of MZI Layout 3. The effective index is modeled as $n_{eff}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2$, with extracted parameters: $n_1 = 2.4444$, $n_2 = -1.1104\,\mu m^{-1}$, and $n_3 = -0.0114\,\mu m^{-2}$. (a) Final numerical fit, (b) Group index extraction, and (c) Initial peak detection.



(a) Experimental data vs. fitted MZI transfer function for Layout 4.

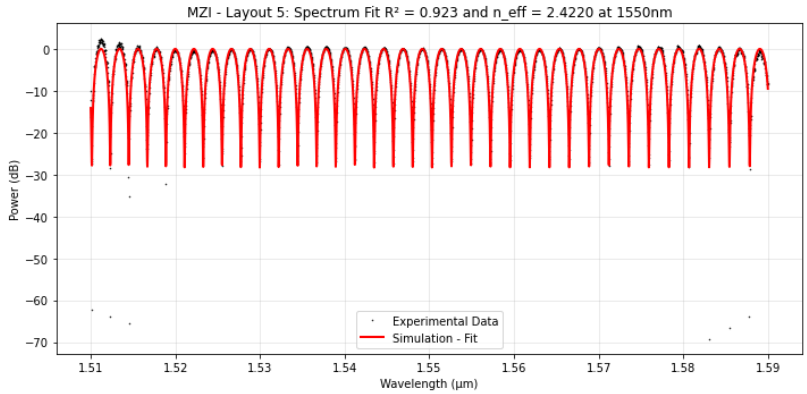


(b) Extracted group index n_g for Layout 4.

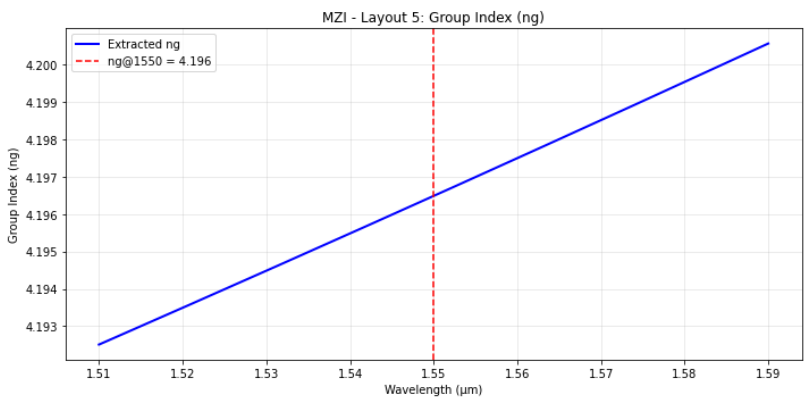


(c) Autocorrelation for Layout 4 used to determine initial FSR.

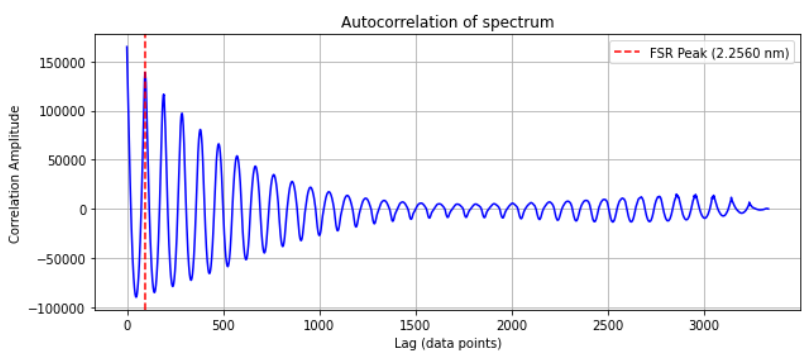
Fig. 13. Full analysis of MZI Layout 4. The effective index is modeled as $n_{eff}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2$, with extracted parameters: $n_1 = 2.4184$, $n_2 = -1.1426\,\mu m^{-1}$, and $n_3 = -0.0510\,\mu m^{-2}$. (a) Final numerical fit, (b) Group index extraction, and (c) Initial peak detection.



(a) Experimental data vs. fitted MZI transfer function for Layout 5.



(b) Extracted group index n_g for Layout 5.



(c) Autocorrelation for Layout 5 used to determine initial FSR.

Fig. 14. Full analysis of MZI Layout 5. The effective index is modeled as $n_{eff}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2$, with extracted parameters: $n_1 = 2.4220$, $n_2 = -1.1448\,\mu m^{-1}$, and $n_3 = -0.0325\,\mu m^{-2}$. (a) Final numerical fit, (b) Group index extraction, and (c) Initial peak detection.

Table 6. Comparison of Predicted vs. Experimental FSR (nm) across Manufacturing Corners for MZI Designs 1–5. Error (%) is calculated as $(FSR_{exp} - FSR_{nom})/FSR_{nom} \times 100$.

Design #	ΔL (µm)	Nominal FSR	Min FSR (C2)	Max FSR (C4)	Exp. FSR (nm)	Error (%)
1	40	14.289	14.092	14.370	**14.298**	+0.06%
2	80	7.144	7.046	7.185	**7.181**	+0.52%
3	160	3.572	3.523	3.593	**3.579**	+0.20%
4	250	2.286	2.255	2.299	**2.308**	+0.96%
5	250	2.286	2.255	2.299	**2.301**	+0.66%

sensitivity of TM modes to silicon layer height and surface roughness—the five TE designs provided robust spectral data. The measured Free Spectral Range (FSR) across all layouts followed the expected $1/\Delta L$ relationship with high precision, ranging from approximately 14.3 nm to 2.3 nm, confirming the fundamental scalability of the delay-line architecture.

Analysis of the extracted effective index (n_{eff}) and group index (n_g) reveals a consistent, high-fidelity match with the numerical simulations. The experimental results across all designs sit comfortably within the predicted manufacturing tolerance boundaries, generally leaning toward the "Fast" Corner 4 (C4). The systematic positive error in FSR, which remained below 1% for all layouts, suggests that the fabricated waveguides are slightly narrower or thinner than the nominal 500×220 nm design. This deviation reduces the group index (n_g), thereby increasing the FSR, but remains well within the acceptable margins for silicon photonics manufacturing.

Finally, the comparison between the straight geometry of Layout 4 and the serpentine format of Layout 5 demonstrates that folding the delay line to save chip area does not compromise performance. Both designs, sharing a ΔL of 250 µm, yielded nearly identical FSR values with a difference of only 0.007 nm ($< 0.3\%$). This indicates that the bend-induced dispersion in the serpentine sections is negligible and that the Taylor expansion model used for n_{eff} remains robust regardless of waveguide curvature. These findings confirm that compact, serpentine MZIs are a reliable solution for high-density integration in photonic integrated circuits.

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